

The goal

You're solving a **first-order linear ODE**:

$$x'(t) + a(t)x(t) = b(t).$$

The annoying part is the $a(t)x(t)$ term — it prevents you from just integrating x' directly.

The integrating factor trick is: **find a multiplier $M(t)$** so that the left-hand side becomes the derivative of a product:

$$\frac{d}{dt}(M(t)x(t)).$$

If we can do that, then the equation becomes “one derivative equals something,” which we can integrate immediately.

How to choose $M(t)$ (derive it, don't memorize it)

Start with the product rule:

$$\frac{d}{dt}(Mx) = Mx' + M'x.$$

We want this to match the left side of the ODE after multiplying the ODE by M :

$$Mx' + Max.$$

Compare the two expressions:

- Product rule gives: $Mx' + M'x$
- We want: $Mx' + (Ma)x$

So we need

$$M'(t)x(t) = M(t)a(t)x(t)$$

for all $x(t)$. Cancel $x(t)$ (the choice of M shouldn't depend on the particular solution):

$$M'(t) = a(t)M(t).$$

That's a separable ODE:

$$\frac{M'}{M} = a(t) \quad \Rightarrow \quad \ln M(t) = \int a(t) dt + C$$

so

$$M(t) = e^{\int a(t) dt}.$$

That's *the* integrating factor — it is chosen exactly so the product rule reproduces your left-hand side.

Once you have $M(t)$, the ODE collapses

Multiply the ODE by $M(t)$:

$$Mx' + Max = Mb.$$

But by construction, $M' = aM$, so

$$Mx' + M'x = Mb \quad \Rightarrow \quad \frac{d}{dt}(Mx) = Mb.$$

Now integrate both sides from 0 to t :

$$M(t)x(t) - M(0)x(0) = \int_0^t M(s)b(s) ds.$$

Solve for $x(t)$:

$$x(t) = \frac{M(0)x(0) + \int_0^t M(s)b(s) ds}{M(t)}.$$

That's the general solution.

A concrete example (so it's not abstract)

Solve:

$$x' + 3x = 6.$$

Here $a(t) = 3$, $b(t) = 6$.

Integrating factor:

$$M(t) = e^{\int 3 dt} = e^{3t}.$$

Multiply the ODE by e^{3t} :

$$e^{3t}x' + 3e^{3t}x = 6e^{3t}.$$

Left side is exactly:

$$\frac{d}{dt}(e^{3t}x) = 6e^{3t}.$$

Integrate:

$$e^{3t}x = \int 6e^{3t} dt = 2e^{3t} + C.$$

Divide by e^{3t} :

$$x(t) = 2 + Ce^{-3t}.$$

You can see the structure: the solution is "steady-state" 2 plus a decaying transient Ce^{-3t} .

The one-sentence intuition

The integrating factor $M(t)$ is chosen to **cancel the drift term** $a(t)x(t)$ by converting the left side into a single total derivative $d(Mx)/dt$.

If you want, paste your OU line again and I'll show the exact one-to-one mapping: $x \leftrightarrow r_t$, $a(t) \leftrightarrow \alpha$, and "integrate dt " becomes "integrate dt and dZ_t " (with Itô handling the product).